

Propositional Logic: Logical Agents, Syntax, Semantics, and Entailment

Lecture 13

A Different Kind of AI Question

Imagine a warehouse robot approaching a charging bay.

What the robot knows

BatteryLow

SmokeDetected $\rightarrow$ BayUnsafe

$\text{BatteryLow} \wedge \neg \text{BayUnsafe} \rightarrow \text{EnterBay}$

The immediate question is not “Which path is shortest?”

What conclusions are justified by what I know?

Key idea

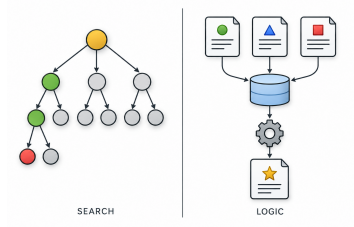
Search reasons over possible actions. Logic reasons over what must be true given an agent's knowledge.



From Search Agents to Logical Agents

Search-oriented agent

1. Represent a state.
2. Generate successor states.
3. Evaluate paths or actions.
4. Choose an action.



Logical agent

1. Represent facts and rules.
2. Add observations to a knowledge base.
3. Infer consequences.
4. Act using those conclusions.

The Knowledge-Based Agent Loop

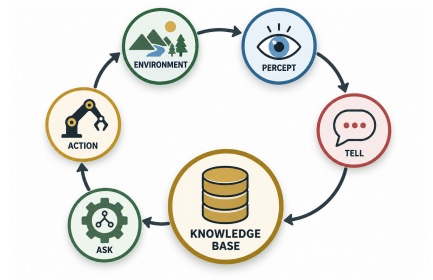
A classical knowledge-based agent repeatedly updates and queries an explicit knowledge base.

1. **Perceive** something about the world.
2. **TELL** the knowledge base what was observed.
3. **ASK** what follows from the current knowledge.
4. Choose an action.
5. Record the action and repeat.

Two separate problems

Representation: What can the agent say?

Inference: What can the agent conclude?



A Knowledge Base Stores More Than Raw Data

A database might store isolated facts:

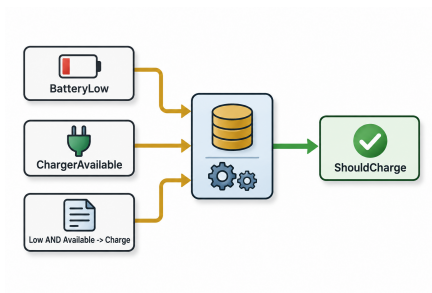
BatteryLow

ChargerAvailable

A knowledge base may also store **rules connecting facts**:

$\text{BatteryLow} \wedge \text{ChargerAvailable} \rightarrow \text{ShouldCharge}$

The sentence ShouldCharge does not need to be written explicitly if it can be derived.



Key idea

Inference turns explicit knowledge into implicit knowledge.

Explicit Knowledge vs. Implicit Knowledge

Explicitly stored

$$P$$
$$P \rightarrow Q$$

These sentences are directly present in the knowledge base.

Implicitly supported

$$Q$$

The agent may be justified in accepting Q , even if Q was never entered directly.

Key idea

A logical agent is valuable because it can derive consequences that were not explicitly enumerated in advance.

Unknown Is Not the Same as False

Suppose the knowledge base contains

$\text{SmokeDetected} \rightarrow \text{BayUnsafe}$

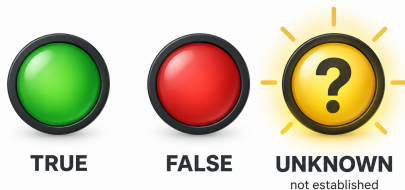
but contains no fact telling us whether smoke is currently detected.

Can the robot conclude

$\neg \text{BayUnsafe}$?

No.

"The KB does not prove P " is different from "the KB proves $\neg P$."

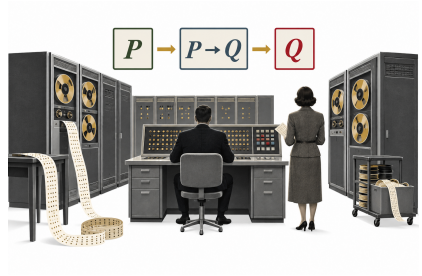


Good Old-Fashioned AI: Make Knowledge Explicit

Much early AI research treated intelligence as the manipulation of explicit symbolic representations.

- ▶ Facts were represented as symbols.
- ▶ Rules specified how symbols could be transformed.
- ▶ Reasoning produced new symbolic statements.
- ▶ The internal reasoning process was often inspectable.

The later label **GOF AI**—“Good Old-Fashioned Artificial Intelligence”—is commonly used for this symbolic tradition.



Key idea

The central wager was that intelligent behavior could emerge from explicit knowledge plus general reasoning procedures.

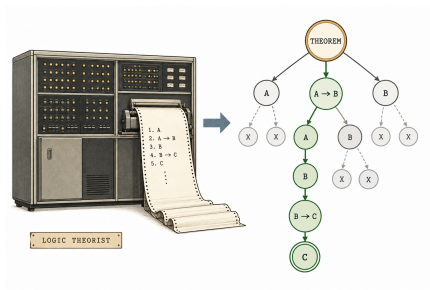
Logic Theorist: Reasoning as Computation

Newell, Simon, and Shaw's *Logic Theorist* in the 1950s demonstrated that a program could search for formal proofs in symbolic logic.

- ▶ The program represented mathematical statements symbolically.
- ▶ It applied legal transformation rules.
- ▶ It searched through possible proof steps.

Notice the recurring AI pattern:

representation + operators + search



Expert Systems: Put the Knowledge in Rules

Later expert systems made the architecture especially explicit:

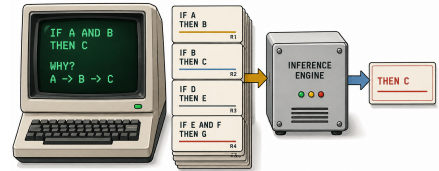
Knowledge base

Domain-specific facts and IF-THEN rules.

Inference engine

A more general mechanism that decides which rules apply and derives conclusions.

This separation is still conceptually important today:
what the system knows is different from **how it reasons**.



Why Symbolic AI Was Powerful—and Why It Was Hard

Strengths

- ▶ explicit, inspectable knowledge;
- ▶ precise rules of reasoning;
- ▶ strong performance in structured domains;
- ▶ conclusions can often be explained as proofs.

Difficulties

- ▶ knowledge engineering is expensive;
- ▶ the real world is messy and incomplete;
- ▶ rule sets can become large and brittle;
- ▶ inference itself can become computationally expensive.

Key idea

Symbolic AI did not disappear; logic remains one of the clearest ways to study representation, reasoning, and guarantees.

Propositional Logic: The Simplest Useful Logic

Definition

A **proposition** is a statement that is either true or false.

Examples:

R = "It is raining."

B = "The battery is low."

U = "The charging bay is unsafe."

Propositional logic treats each atomic statement as a single indivisible symbol. It does **not** yet represent objects, properties, variables, or quantifiers.

Key idea

Propositional logic is limited, but its simplicity makes the mechanics of logical reasoning very clear.

Atomic Symbols Need an Interpretation

The symbol P has no meaning by itself.

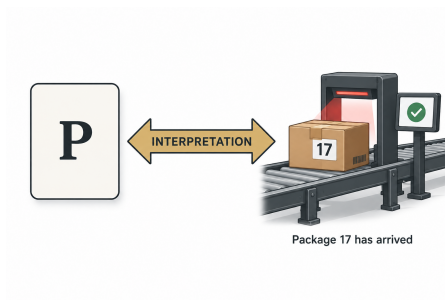
We assign it an interpretation such as

$P = \text{"Package 17 has arrived."}$

Another symbol might mean

$Q = \text{"Robot 2 is available."}$

Logic manipulates the symbols, while the interpretation connects those symbols to a domain.



Building Sentences with Logical Connectives

Symbol	Name	Example
$\neg$	NOT / negation	$\neg P$
$\wedge$	AND / conjunction	$P \wedge Q$
$\vee$	OR / disjunction	$P \vee Q$
$\rightarrow$	implication	$P \rightarrow Q$
$\leftrightarrow$	biconditional	$P \leftrightarrow Q$

Example

$\text{BatteryLow} \wedge \neg \text{BayUnsafe} \rightarrow \text{EnterBay}$

Complex sentences are built recursively from atomic propositions and connectives.

Parentheses Matter

Compare

$$(P \vee Q) \wedge R$$

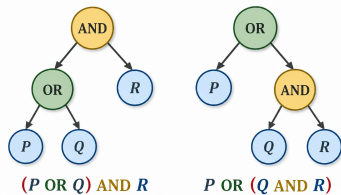
with

$$P \vee (Q \wedge R).$$

These are different sentences and can have different truth values in the same world.

Practical rule

When in doubt, use parentheses. A logic expression should communicate its structure unambiguously to both the reader and the inference system.



A Knowledge Base Is a Set of Sentences

Suppose a delivery robot's KB contains

$$B$$

$$B \rightarrow C$$

$$C \rightarrow M$$

where

- ▶ B : battery is low;
- ▶ C : robot should charge;
- ▶ M : robot should move toward the charger.

Key idea

The knowledge base is not one fact; it is a collection of sentences whose consequences interact.

What Does $P \rightarrow Q$ Actually Say?

An implication

$$P \rightarrow Q$$

states that whenever P is true, Q must also be true.
The implication is violated only by a situation where

$$P = \text{true}, \quad Q = \text{false}.$$

Example:

$$\text{BatteryLow} \rightarrow \text{ShouldCharge}.$$

$P \rightarrow Q$	
$P=T, Q=T$ ✓	$P=T, Q=F$ ✗ VIOLATION
$P=F, Q=T$ ✓	$P=F, Q=F$ ✓

Implication Is Not Causation

The sentence

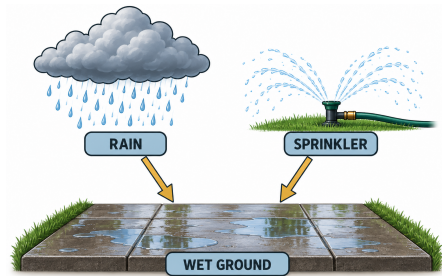
Rain $\rightarrow$ WetGround

does **not** claim that rain is the only possible cause of wet ground.

Sprinklers might also make the ground wet.

Propositional implication expresses a truth constraint:

if the antecedent holds, the consequent must hold.



Key idea

Logical implication describes a relationship among truth values, not a physical mechanism.

Do Not Run the Rule Backward

Suppose the KB contains

$$\text{Rain} \rightarrow \text{WetGround}$$

and we observe

WetGround.

Can we conclude Rain?

No.

The implication only guarantees wet ground when rain occurs. It does not say wet ground can only be caused by rain.

Likewise

From $P \rightarrow Q$ and Q , we cannot generally conclude P .

Quick Check: Which Sentences Say What We Mean?

Let

S = “smoke is detected”, U = “the bay is unsafe.”

Which sentence best represents

“If smoke is detected, then the bay is unsafe”?

1. $S \wedge U$
2. $S \rightarrow U$
3. $U \rightarrow S$
4. $S \leftrightarrow U$

Answer

$S \rightarrow U$. The other formulas make stronger or simply different claims.

Syntax vs. Semantics

Syntax

The legal structure of expressions.

$$(P \wedge Q) \rightarrow R$$

Syntax answers:

Is this a well-formed sentence?

Semantics

The conditions under which a sentence is true or false.

Semantics answers:

What would the world have to be like for this sentence to be true?

Key idea

Logic needs both a formal language and a precise account of what sentences mean.

A Model Is One Possible World

Definition

A **model** assigns a truth value to every proposition symbol under consideration.

With two symbols P and Q , one model might be

$$P = \text{true}, \quad Q = \text{false}.$$

Another model might be

$$P = \text{false}, \quad Q = \text{false}.$$

Each model represents one logically possible configuration of the world at the level of detail captured by those symbols.

Models Grow Exponentially

With n independent proposition symbols, there are

$$2^n$$

possible truth assignments.

Symbols	Possible models
1	2
2	4
3	8
10	1,024

Key idea

Thinking in models gives logic a precise semantics. Next lecture, the number of models becomes a computational problem.

Satisfaction: Does a Model Make a Sentence True?

Suppose model m assigns

$$P = \text{true}, \quad Q = \text{false}.$$

Then

$$m \models P$$

and

$$m \models \neg Q.$$

But

$$m \not\models P \wedge Q.$$

Definition

We write $m \models \alpha$ when model m **satisfies** sentence α .

A Knowledge Base Rules Out Possible Worlds

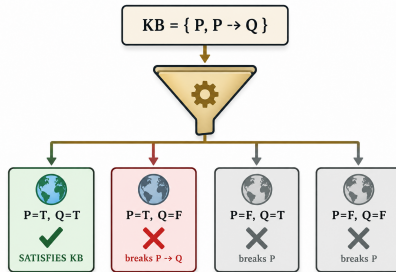
Suppose

$$KB = \{P, P \rightarrow Q\}.$$

Any model in which P is false is incompatible with the first sentence.

Any model in which P is true and Q is false is incompatible with the second sentence.

The models that survive are exactly the worlds consistent with **all** sentences in the KB.



Key idea

Knowledge constrains the set of worlds the agent considers possible.

Entailment

Definition

$$KB \models \alpha$$

means that α is true in **every model** in which the knowledge base is true.

Read it as:

“The knowledge base **entails** α .”

Or more intuitively:

There is no world consistent with what I know in which α is false.

Key idea

Entailment is about necessity relative to the knowledge base, not about whether a conclusion merely seems plausible.

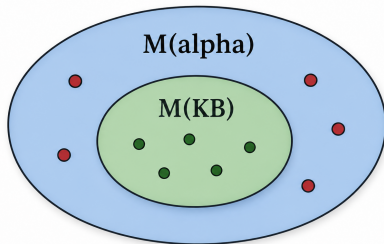
Entailment as Model-Set Containment

Let $M(KB)$ be the set of models satisfying the knowledge base and $M(\alpha)$ the set satisfying α .
Then

$$KB \models \alpha$$

exactly when

$$M(KB) \subseteq M(\alpha).$$



$M(KB)$ subset $M(\alpha)$

Example: Does the KB Entail the Conclusion?

Consider

$$KB = \{\text{BatteryLow}, \text{BatteryLow} \rightarrow \text{ShouldCharge}\}.$$

Query:

ShouldCharge?

Ask the semantic question:

Is there **any** model in which both KB sentences are true but ShouldCharge is false?

There is not. Therefore

$$KB \models \text{ShouldCharge}.$$

Important

Today we are describing **why** the entailment holds. Next lecture we turn that idea into an inference procedure.

Non-Entailment Needs Only One Counterexample

Suppose

$$KB = \{\text{WetGround}\}.$$

Does

$$KB \models \text{Rain}$$

hold?

No. A model with

$$\text{WetGround} = \text{true}, \quad \text{Rain} = \text{false}$$

is perfectly consistent with the KB.

Key idea

To show entailment, every KB-model must support the conclusion. To refute entailment, one countermodel is enough.

Classic Logical-Agent Example: Wumpus World

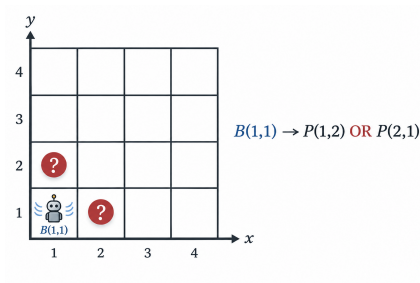
In the classic Wumpus World, an agent receives local percepts such as breezes and stench but must reason about hidden hazards.

A rule might encode an idea like

$$B_{1,1} \rightarrow (P_{1,2} \vee P_{2,1}),$$

meaning a breeze at one square implies a pit in at least one neighboring square.

The agent's job is to combine percepts and rules to determine which locations are safe or dangerous.



Key idea

A logical agent can reason about states of the world it has never directly observed.

What We Have—and What We Still Need

We now have the pieces needed to state a logical reasoning problem:

1. A formal language for writing sentences.
2. A knowledge base containing sentences.
3. Models that define what those sentences mean.
4. Entailment to define what conclusions logically follow.

But there is still a practical question:

How does the computer determine whether $KB \models \alpha$?

Next lecture: Propositional Inference

Logical equivalence, truth tables, model checking, and inference procedures.

Takeaways

- ▶ Knowledge-based agents explicitly represent facts and rules, then derive consequences through inference.
- ▶ Propositional logic builds complex sentences from atomic true/false propositions and logical connectives.
- ▶ **Syntax** determines legal sentence structure; **semantics** determines truth in a model.
- ▶ A knowledge base constrains which possible worlds remain consistent with what the agent knows.
- ▶ $KB \models \alpha$ means α is true in every model satisfying the knowledge base.
- ▶ Entailment defines what logically follows; an inference algorithm tells us how to compute or prove it.

Key idea

Logic gives an AI agent an explicit language for knowledge and a precise standard for when a conclusion is justified.